
A-Worthy Education

H2 Economics 9570 Study Suite

Document 55

Macro CSQ Set C — Model Answers

A-Grade Student-Script Style with Examiner Notes

Contents

Case Study C1 — Model Answers

(a) Two features of South Korea's macroeconomic position [2 marks]

Worked Example

Two features: (i) The current account surplus widened to 4.0% of GDP in 2024, indicating strong export competitiveness; (ii) household debt fell from 105% to 95.1% of GDP between 2022 and 2024, consistent with high interest rates dampening household borrowing.

(b) Effect of KRW depreciation on price and output [6 marks]

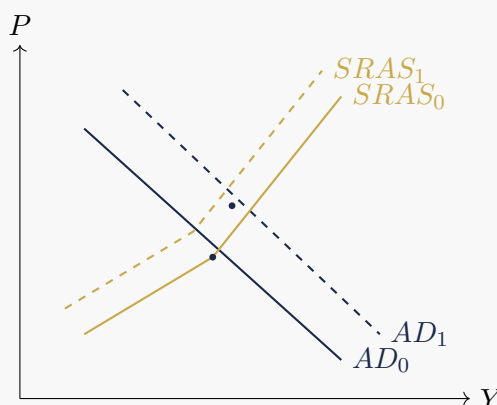
Worked Example

A 22% depreciation of the won raises the won price of imports (the import-price channel) and lowers the dollar price of Korean exports. The transmission has two effects on the domestic economy.

AD channel. Cheaper exports in USD and more expensive imports in KRW improve net exports. AD shifts rightward from AD_0 to AD_1 , raising real output from Y_0 to Y_1 and pushing the price level upward from P_0 to P_1 .

AS channel. Korean producers rely on imported energy, food, and intermediates. The won depreciation raises input costs, shifting SRAS leftward from $SRAS_0$ to $SRAS_1$. This raises the price level further and reduces real output.

The net effect on the price level is unambiguously upward, consistent with the 5.7% inflation peak in 2022. The net effect on real output is ambiguous in theory; in practice, Korean GDP slowed to 1.4% in 2023, suggesting the AS effect dominated in the short run.



(c) Marshall-Lerner and the J-curve [4 marks]

Worked Example

The Marshall-Lerner condition states that a currency depreciation improves the current account only if the sum of the absolute values of the price elasticities of demand for exports and imports exceeds unity: $|PED_X| + |PED_M| > 1$.

In the short run, trade contracts and habitual purchasing patterns make trade volumes inelastic. Although import prices rise immediately in domestic currency, import volumes do not fall proportionately; export volumes also adjust slowly. The current account therefore

worsens before it improves — the J-curve. In the long run, elasticities rise above unity as buyers substitute, and the trade balance improves.

(d) Effectiveness of BOK rate hikes [8 marks]

Worked Example

Effectiveness. The BOK rate increase from 1.25% to 3.50% raises borrowing costs for households and firms, dampening consumption (especially housing-related) and investment. Higher rates also attract capital inflows, supporting the won and reducing imported inflation. The disinflation from 5.7% in mid-2022 to 1.5% by mid-2024 is consistent with policy effectiveness.

Limitations. First, the won depreciated 22% nominal over the same period — suggesting capital flowed toward the higher-yielding US dollar despite Korean rate hikes (Fed rates rose more aggressively). The exchange-rate channel of monetary policy was therefore weakened. Second, the inflation was substantially cost-push (imported energy and food), which monetary policy can only address indirectly. Third, household debt at over 100% of GDP makes the transmission painful — small rate rises produce large debt-service shocks.

Judgement. The BOK rate increase was necessary but not sufficient. Disinflation owes more to the easing of global commodity prices and the unwinding of pandemic supply-chain disruptions than to monetary tightening alone. Policy effectiveness is best described as “a needed contributor to disinflation, but not the primary cause”.

(e) Structural vulnerability of semiconductor reliance [10 marks]

Worked Example

Sources of vulnerability. Semiconductors at 19% of exports concentrates Korea’s external earnings on a single, highly cyclical industry. Memory chip prices fell 35% in 2023, immediately compressing export revenue. The 2023 GDP slowdown to 1.4% — half the historical average — illustrates the macro impact. Beyond cyclicity, Korea faces geopolitical exposure: 60% of memory production is in chips destined for or transiting China; US export controls and CHIPS Act restrictions raise compliance costs and threaten market access.

Mitigating factors. Korea’s two memory champions (Samsung, SK Hynix) together produce 70% of global DRAM, granting near-oligopolistic pricing power. Cyclical downturns have historically resolved within 12–18 months as supply discipline returns. The current account surplus of 4.0% in 2024 indicates the export base is broader than memory alone (autos, shipbuilding, petrochemicals provide diversification).

Structural adjustment. The Korean government has invested heavily in non-memory semiconductors (foundry, design) and in AI-related logic chips, which are less cyclical. The recovery in chip exports in 2024 on AI demand illustrates that the structural pivot is underway.

Judgement. Korea’s semiconductor reliance is a real but partial vulnerability. The cyclicity is genuine, but the duopoly position, the broader manufacturing base, and the ongoing pivot to logic chips mitigate the long-run risk. Korea is structurally more diversified than, for example, an oil-dependent Gulf state, but more concentrated than peer manufacturing economies like Germany or Japan. The vulnerability is best addressed by accelerated diversification rather than reduced semiconductor investment.

Case Study C2 — Model Answers**(a) Two features of India's growth [2 marks]****Worked Example**

(i) India's growth at 8.2% in 2023–24 is the fastest among major economies; (ii) the economy is services-led, with services contributing 53% of GDP versus manufacturing at 17%, well below the Make-in-India target of 25%.

(b) Food inflation versus core CPI [4 marks]**Worked Example**

Food inflation responds primarily to supply-side factors — monsoon variability, crop disease, transportation bottlenecks, and global cereal prices — none of which are sensitive to domestic monetary policy. Core inflation, which excludes food and fuel, reflects demand-side and wage pressures that monetary policy can address through the interest-rate channel. As the RBI tightened rates from 4.00% to 6.50%, core inflation fell from 6.1% to 3.6%, but food inflation rose from 6.7% to 7.5%, reflecting heatwaves and uneven rainfall affecting cereal and pulse crops. The divergence highlights the limits of monetary policy in addressing supply-driven price pressures.

(c) Jobless growth in India [6 marks]**Worked Example**

Concept. Jobless growth refers to a situation in which real GDP rises but employment does not grow commensurately — output expansion is driven by productivity gains, capital deepening, or shifts toward capital-intensive sectors rather than by additional labour absorption.

Evidence for India. Employment elasticity of growth in India fell to 0.1 — every 1% GDP rise produced only 0.1% employment growth — well below historical averages of 0.4–0.5. Female labour force participation at 24% is among Asia's lowest, indicating large untapped labour supply. Yet record capex (32.5% of GDP) is concentrated in physical infrastructure and capital-intensive manufacturing rather than labour-intensive segments.

Assessment. India's growth pattern shows clear features of jobless growth, with the rapid services-led expansion (finance, IT, telecoms) absorbing a small share of the labour force. The case is qualified, however: high-skill services employment is growing rapidly even as headline employment elasticity is low.

Mechanism. The structural problem is a missing middle — manufacturing share is too small to absorb low-skill labour transitioning out of agriculture, while services demand high skills that the education system does not yet supply at scale.

(d) Effectiveness of RBI interest-rate policy [8 marks]**Worked Example**

Effectiveness. The repo-rate hike from 4.00% (early 2022) to 6.50% (end 2023) reduced core CPI from 6.1% to 3.6% — a substantial achievement, consistent with the textbook channel of higher rates dampening demand-side inflation. Capital-account stability was preserved despite the Fed tightening, supporting the rupee within a narrow band.

Limitations. Food prices, accounting for 45% of the CPI basket, are not interest-rate sensitive — RBI tightening did not lower food inflation, which actually rose from 6.7% to 7.5%. India's high informal-sector share (>80% of workers) means the formal credit channel reaches only a minority of the economy. Inflation expectations remain elevated, partly because the politically salient food component remains stubbornly high.

Limiting condition. Effectiveness is contingent on the source of inflation. RBI policy is well-calibrated for demand-driven core inflation but poorly equipped for supply-driven food inflation. The latter requires fiscal interventions (buffer stocks, irrigation investment, agricultural reform).

Judgement. The RBI's policy has been effective on the dimension it controls (core inflation) but cannot fully address the dimension that most affects voters and household real incomes (food inflation). Effectiveness is partial — accurately characterised as “targeted but incomplete”.

(e) Sustainability of India's growth model [10 marks]**Worked Example**

Sustainable features. India's demographic dividend (median age 28, working-age population growing through 2045) provides labour supply for decades. Record public capex (32.5% of GDP) builds infrastructure that raises productivity over the long run. The services-led model exploits comparative advantage in English-language, high-skill output and is less exposed to manufacturing automation than traditional export-led models. FDI inflows reflect investor confidence in the long-run trajectory.

Unsustainable features. Jobless growth means the demographic dividend is not being converted to employment. Female participation at 24% indicates large untapped potential and risks social and political backlash. The services-led model concentrates gains among a narrow skilled segment, exacerbating inequality (Gini rising). Manufacturing under-development means India is increasingly bypassed in global supply chains.

Conditional sustainability. The trajectory is sustainable if and only if (i) manufacturing scales to absorb labour transitioning from agriculture, (ii) female participation rises, and (iii) the education system delivers the skills required by both services and manufacturing. Without these, headline growth may continue but with rising inequality and youth unemployment becoming politically destabilising.

Judgement. India's growth model is sustainable in the headline GDP sense but not yet in the inclusive-development sense. The transition from “fast growth” to “broad-based development” is the critical next decade challenge. The Make-in-India target of 25% manufacturing share and the SkillsFuture-style PMKVY programmes are policy responses, but their scale remains insufficient relative to the labour force size (approximately 470 million workers).

Case Study C3 — Model Answers

(a) Two features of Eurozone recovery [2 marks]

Worked Example

(i) Recovery has been asymmetric — Germany contracted (-0.2%) while Spain grew strongly (3.1%); (ii) southern Eurozone countries (Spain, Italy, Greece) show stronger growth than the historic core (Germany, France), reversing the pre-shock pattern.

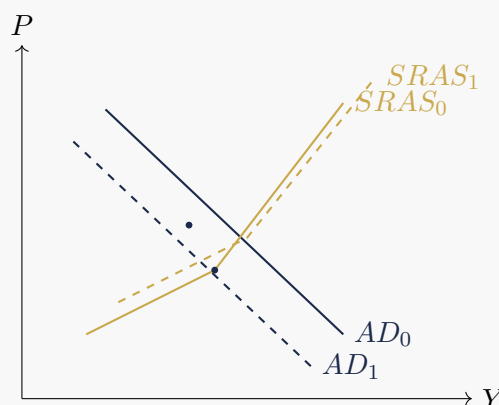
(b) Energy shock — twin effects [6 marks]

Worked Example

An energy price rise affects the Eurozone economy through both AD and AS channels. **AS channel (dominant).** Higher input costs raise marginal cost across all energy-using industries. Marginal social cost rises; the SRAS curve shifts leftward from $SRAS_0$ to $SRAS_1$. The price level rises and real output contracts — the textbook cost-push outcome. Eurozone HICP rose to 10.6% in October 2022, consistent with a large leftward AS shift.

AD channel. Energy is a near-necessity. Rising energy prices reduce real disposable income for households and squeeze profit margins for firms. Both effects reduce consumption and investment, shifting AD leftward from AD_0 to AD_1 . The AD effect lowers the price level (partially offsetting the AS effect) but reinforces the output contraction.

Net effects. The price level rises unambiguously (AS effect dominates AD effect on prices). Real output falls unambiguously (both channels reduce Y). The diagram resembles classic stagflation.



(c) Sharper trade-off for the ECB [6 marks]

Worked Example

A central bank addressing a pure demand-driven inflation faces a moderate trade-off: tighter policy reduces both inflation and output, but ultimately delivers price stability with a temporary output cost.

The ECB, facing a supply-driven (energy-cost) shock, faces a sharper trade-off. Monetary tightening reduces AD but cannot directly reverse the supply contraction. Tightening

therefore worsens the output cost without proportionately reducing the underlying source of inflation.

In addition, the energy shock heterogeneously affected member states. Germany (energy-intensive manufacturing) needed a different stance from Spain (services and tourism). A single ECB rate is necessarily a compromise that is too tight for some members and too loose for others. The trade-off is therefore both temporal (inflation vs. output) and cross-sectional (one member vs. another).

Finally, the ECB's mandate is narrowly price-stability-focused; it cannot offset the asymmetric output damage with member-state-specific fiscal transfers — that would require an EU-level fiscal capacity that does not yet fully exist.

(d) Monetary policy alone — adequacy [8 marks]

Worked Example

Argument that monetary policy alone is inadequate. The shock is supply-side; monetary policy works mainly on demand-side. Tightening would have to be calibrated to deliberately compress AD by enough to offset the price-level rise from the AS shift — implying a severe recession. The deposit-rate rise from -0.50% to 4.00% (450 basis points) is among the largest in ECB history, yet inflation peaked at 10.6% , well above the 2% target. The output cost was substantial (Germany in two-year contraction). Monetary policy can address second-round effects (wage-price spiral, inflation expectations) but not the first-round price rise from cost-push.

Required complementary policies.

- *Energy supply diversification:* Eurozone reduced Russian gas dependence from 40% to under 10% within 18 months — the underlying cause of the shock was addressed by supply policy, not monetary policy.
- *Fiscal cushioning:* Member-state energy subsidies and price caps protected real incomes, slowing the second-round wage response.
- *Supply-side investment:* Acceleration of renewables and grid investment is reducing future shock exposure.

Judgement. Monetary policy was a necessary component of the response but clearly not sufficient. The disinflation from 10.6% to under 3% reflects (i) global energy normalisation, (ii) fiscal cushioning, and (iii) monetary tightening — in roughly that order of importance. A monetary-only response would have produced a deeper recession with similar disinflation outcomes.

(e) Single monetary policy — optimal for members? [8 marks]

Worked Example

Evidence of sub-optimality. The 2024 cross-section shows striking divergence. Germany at -0.2% growth and 2.5% inflation needs lower rates. Spain at 3.1% growth and 2.9% inflation does not. A single ECB rate cannot simultaneously be the right level for both. Unemployment ranges from 3.5% (Germany) to 11.4% (Spain) — the labour market slack profile is highly heterogeneous.

Theory. Optimum currency area (OCA) theory (Mundell, 1961) identifies four criteria:

labour mobility, fiscal transfers, similar shocks, and price/wage flexibility. The Eurozone fails on at least three: labour mobility is limited by language and culture; fiscal transfers between members are constrained by Treaty rules; shocks are increasingly asymmetric (energy intensity, manufacturing share, household debt structure).

Counter-evidence. The Eurozone has survived a decade of stress (Greek crisis, Brexit, energy shock, COVID) without member exits. Convergence on inflation in 2024 (2.4–3.0% range across major members) is tighter than it was in 2022 (3–13% range). The ECB’s targeted longer-term refinancing operations (TLTROs) and Outright Monetary Transactions (OMT) provide some country-specific flexibility within the single-rate framework.

Judgement. The single monetary policy is sub-optimal for individual members in the textbook OCA sense, but it is more workable in practice than OCA theory predicts because (i) member-state fiscal policy retains autonomy, (ii) the ECB has developed unconventional country-targeted tools, and (iii) the political cost of exit is prohibitive. Sub-optimal is not the same as unsustainable.

Case Study C4 — Model Answers

(a) Features of Australia's external sector [2 marks]

Worked Example

(i) Three commodities (iron ore, coal, LNG) account for 51% of goods exports — high concentration; (ii) mining contributes 14% of GDP but only 2.6% of employment, indicating an extremely capital-intensive sector.

(b) Resource curse [4 marks]

Worked Example

The resource curse refers to the paradoxical observation that resource-rich economies often experience weaker long-run development than resource-poor economies. Two channels operate.

Dutch disease. The commodity export boom appreciates the currency, raising the relative price of non-tradeables and reducing competitiveness in non-mining tradeables. Australia's manufacturing share fell from 14% (1990) to 5% (2024). The Australian dollar's correlation with iron ore prices is a textbook Dutch disease symptom.

Volatility and investment crowding. Mining is highly cyclical; the Australian terms of trade swings 30–50% between booms and busts. Fiscal revenues swing similarly, complicating long-run public-investment planning.

(c) Fall in iron ore prices — terms of trade and CA [6 marks]

Worked Example

Terms of trade. The terms of trade (TOT) is the ratio of an export price index to an import price index. A fall in iron ore prices, with import prices unchanged, lowers the TOT directly. Each unit of Australian iron ore now buys fewer imports. Cumulatively, a 20% fall in iron ore (from US\$229 to US\$117/tonne since 2021) corresponds to a significant TOT contraction.

Current account. The current account records the trade balance, primary income and secondary income. The fall in iron ore prices reduces export earnings, ceteris paribus. With imports unchanged, the trade balance worsens. Multinational mining firms also remit profits — primary income outflows fall slightly, offsetting part of the trade deterioration. Net effect: current account narrows.

The 2024 current account at 1.7% of GDP is well below the 2021 peak of approximately 3%, consistent with the iron-ore price normalisation.

Magnitude. Australia's current account is sensitive to commodity prices because (i) 51% of exports are commodity-priced and (ii) volume responses are slow (mine capacity is locked in for years). The CA elasticity to commodity prices is approximately 0.5.

(d) Fiscal response to commodity shock [8 marks]**Worked Example**

Counter-cyclical fiscal smoothing. The textbook response is to run surpluses during commodity booms (when the economy is overheating and revenue is windfall) and deficits during busts. Norway's Government Pension Fund Global is the canonical example, accumulating commodity windfalls into a sovereign wealth vehicle for long-run intergenerational transfer.

Application to Australia. Australia's net debt at 19.7% of GDP is low by OECD standards, providing fiscal headroom. The 2024 federal budget shifted to a small surplus despite GDP slowing to 1.4%, reflecting deliberate fiscal restraint.

Alternative — automatic stabilisers. Allowing tax revenue to fall automatically with commodity prices (without raising tax rates) lets the budget cushion the shock. This is less disciplined than a formal sovereign wealth fund but politically simpler.

Limitations. Australia lacks a Norwegian-style commodity-revenue savings vehicle. State governments depend on royalties, complicating fiscal coordination. The mining boom of 2010–2014 generated revenue that was largely spent rather than saved, leaving Australia more exposed to subsequent downturns than it needed to be.

Judgement. Counter-cyclical fiscal policy is the appropriate response, but its effectiveness depends on (i) institutional pre-commitment (a stabilisation fund), (ii) political discipline during booms, and (iii) coordination between federal and state governments. Australia partially achieves (i)–(iii) but underperforms the Norwegian benchmark.

(e) Diversification away from commodity dependence [10 marks]**Worked Example**

Case for diversification. (i) Volatility — commodity price cycles produce GDP and current-account swings that complicate macro management. (ii) Long-run demand outlook — the global energy transition reduces demand for coal and may slow steel demand (and therefore iron ore). (iii) Dutch disease — sustained commodity revenue keeps the currency strong, eroding non-mining tradables (manufacturing already shrunk to 5%). (iv) Employment — mining provides only 2.6% of jobs despite 14% of GDP, limiting its labour-market contribution.

Case against forced diversification. (i) Comparative advantage — Australia's geology and infrastructure make it among the world's lowest-cost commodity producers; diversification means moving toward sectors where the country has no comparative advantage. (ii) Past attempts — manufacturing assistance programmes (car industry subsidies, 1980s–2010s) failed to create durable competitive industries. (iii) Services strength — Australia is already diversified into services (60%+ of GDP), tourism, education exports, and financial services, which complement rather than replace mining. (iv) Lithium and rare earths — the energy transition raises demand for Australia's other commodities; the sector is naturally rotating within the commodity basket.

Conditional judgement. Active diversification *within* commodities (toward critical minerals, hydrogen, processed metals) is well-founded and already underway. Forced diversification *away from* commodities is harder to justify on comparative-advantage grounds. The right policy is to (i) maintain a sovereign-wealth-style fiscal buffer for commodity revenue, (ii) invest mineral rent in skills, R&D, and infrastructure that benefits non-mining sectors, and (iii) avoid heavy-handed industrial policy that crowds out market-determined comparative advantage.

Final view. Australia should not aim to reduce its commodity share; it should aim to reduce the macroeconomic volatility that commodity dependence creates. These are different policy objectives and call for different instruments.

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